

### Advanced (Habanero)

**Q1.** What is  $2\sqrt{3} + 3\sqrt{3}$ ?

- (A)  $5\sqrt{3}$
- (B)  $\sqrt{3}$
- (C)  $6\sqrt{3}$
- (D)  $4\sqrt{3}$
- (E)  $3\sqrt{2}$

**Q2.** Simplify  $\sqrt{12} - \sqrt{3}$

- (A)  $\sqrt{9}$
- (B)  $2\sqrt{3}$
- (C)  $\sqrt{12} - \sqrt{3}$
- (D)  $\sqrt{15}$
- (E)  $\sqrt{9} - \sqrt{3}$

**Q3.** What is  $\sqrt{8} \cdot \sqrt{2}$ ?

- (A) 4
- (B)  $\sqrt{16}$
- (C)  $2\sqrt{2}$
- (D)  $\sqrt{4}$
- (E) 8

**Q4.** Simplify  $\sqrt{20}$

- (A)  $2\sqrt{5}$
- (B)  $\sqrt{4} \cdot \sqrt{5}$
- (C)  $\sqrt{2} \cdot \sqrt{10}$
- (D)  $4\sqrt{5}$
- (E)  $\sqrt{100}$

**Q5.** Simplify  $\frac{\sqrt{24}}{\sqrt{6}}$

- (A)  $\sqrt{4}$
- (B) 2
- (C)  $\sqrt{2}$
- (D)  $\sqrt{3}$
- (E)  $\frac{\sqrt{24}}{\sqrt{6}}$

**Q6.** Simplify  $\sqrt{50}$

- (A)  $\sqrt{25} \cdot \sqrt{2}$
- (B)  $5\sqrt{2}$
- (C)  $25\sqrt{2}$
- (D)  $10\sqrt{5}$
- (E)  $\sqrt{10}$

**Q7.** Simplify  $\sqrt{75}$

- (A)  $5\sqrt{3}$
- (B)  $\sqrt{25} \cdot \sqrt{3}$
- (C)  $25\sqrt{3}$
- (D)  $10\sqrt{3}$
- (E)  $\sqrt{3}$

**Q8.** What is  $(\sqrt{5})^2$ ?

- (A) 25
- (B) 5
- (C) 10
- (D)  $\sqrt{25}$
- (E)  $\sqrt{10}$

#### Open Ended Questions (FRQ)

**Q9.** Simplify  $\sqrt{32} + \sqrt{8}$  and show your work

**Q10.** Simplify  $(\sqrt{6} + \sqrt{2})(\sqrt{6} - \sqrt{2})$  and show your work

#### Chef's Tips

- Emphasize the importance of simplifying radicals
- Make sure students understand how to add, subtract, multiply, and divide radicals
- Encourage students to show their work for free response questions